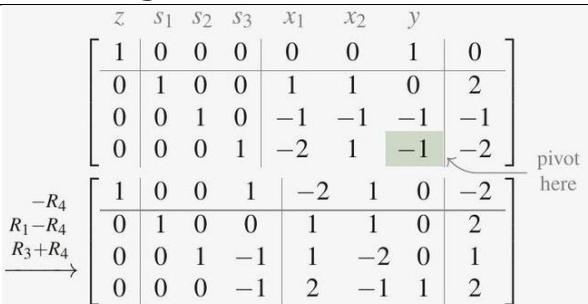
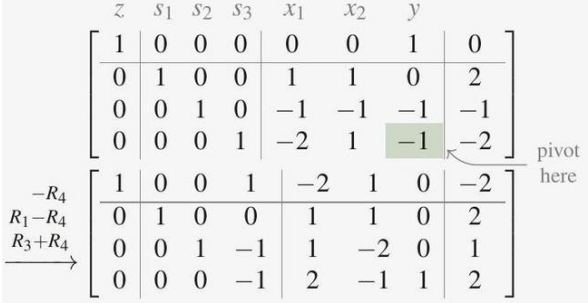
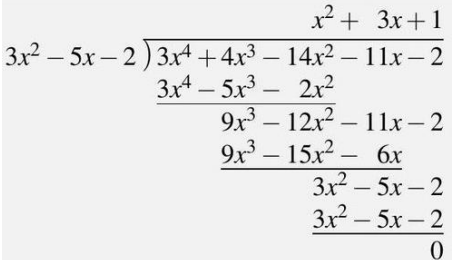


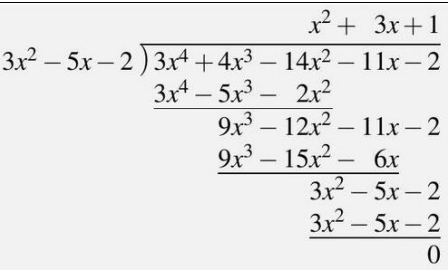
johnston-linear-matrix-algebra: residuals

equations measured 2026-09-15T13:50:28Z against the model of 2026-09-15

Corrected (6)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
johnston-linear-matrix-algebra_E00006 (was)	22	■ 0.141	$\begin{array}{lrl} & \mathbf{x}+2\mathbf{v}+2\mathbf{w} & \\ \Rightarrow & 4\mathbf{x} & \\ \Rightarrow & \mathbf{x} & \\ \Rightarrow & \text{(divide both sides by 4)} & \end{array}$	$\begin{array}{lrl} & \mathbf{x}+2\mathbf{v}+2\mathbf{w} & \\ \Rightarrow & 4\mathbf{x} & \\ \Rightarrow & \mathbf{x} & \\ \Rightarrow & \text{(divide both sides by 4)} & \end{array}$	$\begin{array}{lrl} & \mathbf{x}+2\mathbf{v}+2\mathbf{w} & \\ \Rightarrow & 4\mathbf{x} & \\ \Rightarrow & \mathbf{x} & \\ \Rightarrow & \text{(divide both sides by 4)} & \end{array}$
johnston-linear-matrix-algebra_E00006 (now)	22	—	$\begin{array}{lrl} & \mathbf{x}+2\mathbf{v}+2\mathbf{w} & \\ \Rightarrow & 4\mathbf{x} & \\ \Rightarrow & \mathbf{x} & \\ \Rightarrow & \text{(divide both sides by 4)} & \end{array}$	$\begin{array}{lrl} & \mathbf{x}+2\mathbf{v}+2\mathbf{w} & \\ \Rightarrow & 4\mathbf{x} & \\ \Rightarrow & \mathbf{x} & \\ \Rightarrow & \text{(divide both sides by 4)} & \end{array}$	$\begin{array}{lrl} & \mathbf{x}+2\mathbf{v}+2\mathbf{w} & \\ \Rightarrow & 4\mathbf{x} & \\ \Rightarrow & \mathbf{x} & \\ \Rightarrow & \text{(divide both sides by 4)} & \end{array}$
basis: inferred / ink					
johnston-linear-matrix-algebra_E00014 (was)	27	■ 0.071	$\left(\begin{array}{l} \mathbf{v}+\mathbf{w} \\ \mathbf{x}+\mathbf{y} \end{array}\right) \cdot \left(\begin{array}{l} \mathbf{v}+\mathbf{w} \\ \mathbf{x}+\mathbf{y} \end{array}\right) = \mathbf{v} \cdot \mathbf{v} + \mathbf{w} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{y} + \mathbf{w} \cdot \mathbf{x}$	$\left(\begin{array}{l} \mathbf{v}+\mathbf{w} \\ \mathbf{x}+\mathbf{y} \end{array}\right) \cdot \left(\begin{array}{l} \mathbf{v}+\mathbf{w} \\ \mathbf{x}+\mathbf{y} \end{array}\right) = \mathbf{v} \cdot \mathbf{v} + \mathbf{w} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{y} + \mathbf{w} \cdot \mathbf{x}$	$\begin{array}{lrl} & \mathbf{v}+\mathbf{w} & \\ \Rightarrow & \mathbf{x}+\mathbf{y} & \\ \Rightarrow & \mathbf{v}+\mathbf{w} & \\ \Rightarrow & \mathbf{x}+\mathbf{y} & \\ \Rightarrow & \mathbf{v}+\mathbf{w} & \\ \Rightarrow & \mathbf{x}+\mathbf{y} & \end{array}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
johnston-linear-matrix-algebra- EQ0528 (was)	212	■0.001	$\begin{array}{r} \leftarrow \begin{array}{c} z \\ R_1 \end{array} \rightarrow \begin{array}{c} S_2 \quad S_3 \\ R_3 + R_4 \\ 0 \\ 0 \\ 0 \end{array} \left \begin{array}{cccccc} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{array} \right \begin{array}{c} \left[\begin{array}{ccc ccc} 1 & 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 & -2 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 1 \end{array} \right] \\ 2 \\ \text{pivot} \end{array} \end{array}$		
johnston-linear-matrix-algebra- EQ0528 (now)	212	—	$\begin{array}{l} \rightarrow \begin{array}{c} z \\ R_1 \end{array} \rightarrow \begin{array}{c} S_2 \quad S_3 \\ R_3 + R_4 \\ R_4 + R_4 \end{array} \left[\begin{array}{ccc ccc} 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 1 & -2 & 1 & -2 \end{array} \right] \curvearrowright \text{pivot here} \xrightarrow{\begin{array}{c} -R_4 \\ R_1 - R_4 \\ R_3 + R_4 \end{array}} \left[\begin{array}{ccc ccc} 1 & 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 & -2 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 1 \end{array} \right] \end{array}$		
basis: inferred / ink					
johnston-linear-matrix-algebra- EQ1276 (was)	427	■0.022	$\left(3x^2 - 5x - 2 \right) \longdiv { \frac { x ^ { 2 } } { 3x^4 + 4x^3 - 14x^2 - 11x - 2 } , \frac { 3x^4 - 5x^3 - 2x^2 } { 9x^3 - 12x^2 - 11x - 2 } , \frac { 9x^3 - 15x^2 - 6x } { 3x^2 - 5x - 2 } , \frac { 3x^2 - 5x - 2 } { 0 } }$		
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
johnston-linear-matrix-atlas E01276 (now)	427	—	$\begin{array}{r} x^2+3x+1\backslash \\ \backslash 3x^2-5x-2\overline{)3x^4+4x^3-14x^2-11x-2}\backslash\backslash \\ \underline{3x^4-5x^3-2x^2}\backslash\backslash \\ 9x^3-12x^2-11x-2\backslash\backslash \underline{9x^3-15x^2-6x}\backslash\backslash \\ \backslash\backslash 3x^2-5x-2\backslash\backslash \underline{3x^2-5x-2}\backslash\backslash 0 \end{array}$	$3x^2-5x-2\overline{)3x^4+4x^3-14x^2-11x-2}$ $\underline{3x^4-5x^3-2x^2}$ $9x^3-12x^2-11x-2$ $\underline{9x^3-15x^2-6x}$ $3x^2-5x-2$ $\underline{3x^2-5x-2}$ 0	
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Unresolved

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-E00114 [conf 0.025]	63	■ 0.025 ●	<pre>\begin{figure} \[R^{\theta}\left(\mathbf{e}_{2}\right)=(-\sin (\theta), \cos (\theta)) \overbrace{\cos (\theta)^{-\sin (\theta)}}{\mathbf{e}_{1}^{\mathbf{e}_{2}}} \times \text { - } \underbrace{\cos (\theta)}_{R^{\theta}\left(\mathbf{e}_{1}\right)} \left(\mathbf{e}_{1}\right)=(\cos (\theta), \sin (\theta)) \]</pre> \captionsetup{labelformat=empty} \caption{Figure 1.19: (R^{θ}) rotates the standard basis vectors $(\mathbf{e}_1)$ and $(\mathbf{e}_2)$ to $(R^{\theta}\left(\mathbf{e}_1\right))=(\cos (\theta), \sin (\theta))$ and $(R^{\theta}\left(\mathbf{e}_2\right))=(-\sin (\theta), \cos (\theta))$, respectively.}	(not rendered)	
johnston-linear-matrix-algebra-E00439	170	■ 0.131 ●	<pre>\begin{figure} \[\begin{aligned} {[A \mid I]} &= \left[\begin{array}{cccccccc cccc} * & * & * & * & * & * & * & * & 1 & 0 & 0 & 0 \\ * & * & * & * & * & * & * & * & 0 & 1 & 0 & 0 \\ * & * & * & * & * & * & * & * & 0 & 0 & 1 & 0 \\ * & * & * & * & * & * & * & * & 0 & 0 & 0 & 1 \end{array} \right] \\ \xrightarrow{\text{row-reduce}} & \left[\begin{array}{cccccccc cccc} \star & * & * & * & * & * & * & * & * & * & * & * \\ 0 & 0 & \star & * & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \star & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & * & * & * & * \end{array} \right] = [R \mid E] \end{aligned}</pre> \captionsetup{labelformat=empty} \caption{Figure 2.22: After we row-reduce $([A \mid I])$ to a row echelon form $([R \mid E])$, we can immediately read off bases of $(\text{range}(A^T))$ and $(\text{null}(A^T))$ from the rows of that matrix.}	(not rendered)	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra EQ8575	224	0.324 C +93	<pre> \begin{figure} \[\left.A=\left[\begin{array}{llllll} * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \end{array}\right]=\left[\begin{array}{lll} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{array}\right] \left[\begin{array}{llllll} * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \end{array}\right] \] \captionsetup{labelformat=empty} \caption{Figure 2.39: Theorem 2.C. 1 says that every matrix can be written as a product of a tall and skinny matrix and a short and fat matrix. The rank determines exactly how skinny and short the matrices in the product can be (as shown here, \((A)\) has rank 3, so it can be written as a product of a matrix \((C)\) with 3 columns and a matrix \((R)\) with 3 rows).} \end{figure} </pre>	(not rendered)	The diagram illustrates the rank decomposition of a matrix A. On the left, matrix A is shown as a 10x6 grid of asterisks. This is followed by an equals sign and two matrices: a 10x3 matrix C (tall and skinny) and a 3x6 matrix R (short and fat), both also represented by asterisks. An arrow points from the text $\text{rank}(A) = 3$ to the three columns of matrix C, indicating that the rank of A is equal to the number of columns in C.
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Flagged, not acted on

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra EQ0884	318	■ 0.451 ● C +38	$\begin{aligned} \operatorname{rank}(A) &= \operatorname{rank}(B) = 3, \text{ \& } \operatorname{tr}(A) \text{ \& } \operatorname{tr}(B) = 0, \dots \\ \operatorname{rank}(B) &= 3, \text{ \& } \operatorname{tr}(A) \text{ \& } \operatorname{tr}(B) = 0, \dots \\ \operatorname{tr}(B) &= 0, \dots \end{aligned}$	$\operatorname{rank}(A) = \operatorname{rank}(B) = 3, \quad \operatorname{tr}(A) = \operatorname{tr}(B) = 0, \dots$	$\operatorname{rank}(A) = \operatorname{rank}(B) = 3, \quad \operatorname{tr}(A) = \operatorname{tr}(B) = 0, \quad \det(A) = \det(B) = 2, \quad \text{and} \quad p_A(\lambda) = p_B(\lambda) = -\lambda^3 + 3\lambda + 2.$
johnston-linear-matrix-algebra EQ0793	293	■ 0.462 ● C +130	$\left[\begin{array}{cccc} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ 0 & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & a_{n,2} & \cdots & a_{n,n} \end{array} \right] \quad \text{has row echelon form} \quad \left[\begin{array}{c ccc} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ \hline 0 & & & \\ \vdots & & & \\ 0 & & & R \end{array} \right],$	$\left[\begin{array}{c ccc} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ \hline 0 & & & \\ \vdots & & & \\ 0 & & & R \end{array} \right],$	$\left[\begin{array}{c ccc} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ \hline 0 & & & \\ \vdots & & & \\ 0 & & & R \end{array} \right],$
johnston-linear-matrix-algebra EQ0040	39	■ 0.479 ● C +817	$\begin{aligned} A B &= \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \left[\begin{array}{lll} 5 & 6 & 7 \\ 8 & 9 & 10 \end{array} \right] \\ &= \left[\begin{array}{lll} (1,2) \cdot (5,8) & (1,2) \cdot (6,9) & (1,2) \cdot (7,10) \\ (3,4) \cdot (5,8) & (3,4) \cdot (6,9) & (3,4) \cdot (7,10) \end{array} \right] \\ &= \left[\begin{array}{lll} 1 \times 5 + 2 \times 8 & 1 \times 6 + 2 \times 9 & 1 \times 7 + 2 \times 10 \\ 3 \times 5 + 4 \times 8 & 3 \times 6 + 4 \times 9 & 3 \times 7 + 4 \times 10 \end{array} \right] \\ &= \left[\begin{array}{lll} 21 & 24 & 27 \\ 47 & 54 & 61 \end{array} \right]. \end{aligned}$	$\begin{aligned} AB &= \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \left[\begin{array}{lll} 5 & 6 & 7 \\ 8 & 9 & 10 \end{array} \right] \\ &= \left[\begin{array}{lll} (1,2) \cdot (5,8) & (1,2) \cdot (6,9) & (1,2) \cdot (7,10) \\ (3,4) \cdot (5,8) & (3,4) \cdot (6,9) & (3,4) \cdot (7,10) \end{array} \right] \\ &= \left[\begin{array}{lll} 1 \times 5 + 2 \times 8 & 1 \times 6 + 2 \times 9 & 1 \times 7 + 2 \times 10 \\ 3 \times 5 + 4 \times 8 & 3 \times 6 + 4 \times 9 & 3 \times 7 + 4 \times 10 \end{array} \right] \\ &= \left[\begin{array}{lll} 21 & 24 & 27 \\ 47 & 54 & 61 \end{array} \right]. \end{aligned}$	$\begin{aligned} AB &= \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \left[\begin{array}{lll} 5 & 6 & 7 \\ 8 & 9 & 10 \end{array} \right] \\ &= \left[\begin{array}{lll} (1,2) \cdot (5,8) & (1,2) \cdot (6,9) & (1,2) \cdot (7,10) \\ (3,4) \cdot (5,8) & (3,4) \cdot (6,9) & (3,4) \cdot (7,10) \end{array} \right] \\ &= \left[\begin{array}{lll} 1 \times 5 + 2 \times 8 & 1 \times 6 + 2 \times 9 & 1 \times 7 + 2 \times 10 \\ 3 \times 5 + 4 \times 8 & 3 \times 6 + 4 \times 9 & 3 \times 7 + 4 \times 10 \end{array} \right] \\ &= \left[\begin{array}{lll} 21 & 24 & 27 \\ 47 & 54 & 61 \end{array} \right]. \end{aligned}$
johnston-linear-matrix-algebra EQ0073	48	■ 0.493 ● C +396	$\begin{aligned} A B &= \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \left[\begin{array}{lll} -1 & 1 & 1 \\ 0 & 1 & 0 \end{array} \right] \\ &= \left[\begin{array}{lll} (1,2) \cdot (-1,0) & (1,2) \cdot (1,1) & (1,2) \cdot (1,0) \\ (3,4) \cdot (-1,0) & (3,4) \cdot (1,1) & (3,4) \cdot (1,0) \end{array} \right] \\ &= \left[\begin{array}{lll} -1 & 3 & 1 \\ -3 & 7 & 3 \end{array} \right]. \end{aligned}$	$\begin{aligned} AB &= \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \left[\begin{array}{lll} -1 & 1 & 1 \\ 0 & 1 & 0 \end{array} \right] \\ &= \left[\begin{array}{lll} (1,2) \cdot (-1,0) & (1,2) \cdot (1,1) & (1,2) \cdot (1,0) \\ (3,4) \cdot (-1,0) & (3,4) \cdot (1,1) & (3,4) \cdot (1,0) \end{array} \right] \\ &= \left[\begin{array}{lll} -1 & 3 & 1 \\ -3 & 7 & 3 \end{array} \right]. \end{aligned}$	$\begin{aligned} AB &= \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \left[\begin{array}{lll} -1 & 1 & 1 \\ 0 & 1 & 0 \end{array} \right] \\ &= \left[\begin{array}{lll} (1,2) \cdot (-1,0) & (1,2) \cdot (1,1) & (1,2) \cdot (1,0) \\ (3,4) \cdot (-1,0) & (3,4) \cdot (1,1) & (3,4) \cdot (1,0) \end{array} \right] \\ &= \left[\begin{array}{lll} -1 & 3 & 1 \\ -3 & 7 & 3 \end{array} \right]. \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra EQ0635 (0)	243	■0.508 ●C +27	E=\left[\begin{array}{ccccc} 0 & 0 & \star & 0 & 0 \\ 0 & 0 & \star & \star & 0 \\ \star & 0 & \star & \star & 0 \\ \star & 0 & \star & \star & \star \\ \star & \star & \star & \star & \star \end{array}\right] \text{ then } P^T=\left[\begin{array}{ccccc} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{array}\right] .	$E = \begin{bmatrix} 0 & 0 & \star & 0 & 0 \\ 0 & 0 & \star & \star & 0 \\ \star & 0 & \star & \star & 0 \\ \star & 0 & \star & \star & \star \\ \star & \star & \star & \star & \star \end{bmatrix}$ then $P^T = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}$.	
johnston-linear-matrix-algebra EQ0214	101	■0.533 ●C +146	\left[\begin{array}{llll} \star & \star & \star & \star \\ 0 & 0 & \star & \star \\ 0 & 0 & 0 & \star \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \text{and} \quad \left[\begin{array}{cccccccccccc} 0 & 0 & \star & \star & \star & \star & \star & \star & \star & \star & \star & \star \\ 0 & 0 & 0 & \star & \star & \star & \star & \star & \star & \star & \star & \star \\ 0 & 0 & 0 & 0 & 0 & \star & \star & \star & \star & \star & \star & \star \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \star & \star & \star & \star \end{array}\right] .	$\begin{bmatrix} \star & \star & \star & \star \\ 0 & 0 & \star & \star \\ 0 & 0 & 0 & \star \\ 0 & 0 & 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 & \star & \star & \star & \star & \star & \star & \star & \star & \star & \star \\ 0 & 0 & 0 & \star & \star & \star & \star & \star & \star & \star & \star & \star \\ 0 & 0 & 0 & 0 & 0 & \star & \star & \star & \star & \star & \star & \star \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \star & \star & \star & \star \end{bmatrix}$.	
johnston-linear-matrix-algebra EQ0331 (0)	130	■0.564 ●C +20	\begin{array}{r} \left[\begin{array}{lll lll} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 1 & 2 & 4 \\ 1 & 3 & 9 & 1 & 3 & 9 \end{array}\right] \xrightarrow{\substack{R_2-R_1 \\ R_3-R_1}} \left[\begin{array}{lll lll} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 2 & 1 & -2 & 1 \end{array}\right] \\ \xrightarrow{\frac{1}{2}R_3} \left[\begin{array}{lll lll} 1 & 0 & -2 & 2 & -1 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1/2 & -1 & 1/2 \end{array}\right] \\ \xrightarrow{R_1+2R_3} \left[\begin{array}{lll lll} 1 & 0 & 0 & 3 & -3 & 1 \\ 0 & 1 & 0 & -5/2 & 4 & -3/2 \\ 0 & 0 & 1 & 1/2 & -1 & 1/2 \end{array}\right] . \end{array}	$\left[\begin{array}{lll lll} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 4 & 0 & 1 & 0 \\ 1 & 3 & 9 & 0 & 0 & 1 \end{array}\right] \xrightarrow{\substack{R_2-R_1 \\ R_3-R_1}} \left[\begin{array}{lll lll} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 2 & 8 & -1 & 0 & 1 \end{array}\right] \\ \xrightarrow{\substack{R_1-R_2 \\ R_3-2R_2}} \left[\begin{array}{lll lll} 1 & 0 & -2 & 2 & -1 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 2 & 1 & -2 & 1 \end{array}\right] \\ \xrightarrow{\frac{1}{2}R_3} \left[\begin{array}{lll lll} 1 & 0 & -2 & 2 & -1 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1/2 & -1 & 1/2 \end{array}\right] \\ \xrightarrow{\substack{R_1+2R_3 \\ R_2-3R_3}} \left[\begin{array}{lll lll} 1 & 0 & 0 & 3 & -3 & 1 \\ 0 & 1 & 0 & -5/2 & 4 & -3/2 \\ 0 & 0 & 1 & 1/2 & -1 & 1/2 \end{array}\right] .$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-EQ0519	207	0.720 C +175	$\begin{gathered} \left[\begin{array}{cccc ccc} 1 & 0 & 2 & 0 & 0 & -3 & 3 & 4 \\ 0 & 1 & 3 & 0 & 0 & -3 & 5 & 7 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 2 & -2 & 5 \end{array} \right] \\ 5/2 \text{ is smallest (and only) positive ratio} \end{gathered}$		
johnston-linear-matrix-algebra-EQ0210	99	0.827 C +26	$\begin{gathered} \left[\begin{array}{cccc ccc} 1 & 3 & -2 & 5 & 1 & 3 & -2 & 5 \\ 1 & 5 & -8 & 9 & 2 & 4 & 5 & 12 \\ 2 & 4 & 5 & 12 & 1 & 3 & -2 & 5 \end{array} \right] \\ \xrightarrow{R_2-R_1} \left[\begin{array}{cccc ccc} 1 & 3 & -2 & 5 & 0 & 2 & -6 & 4 \\ 1 & 5 & -8 & 9 & 2 & 4 & 5 & 12 \\ 2 & 4 & 5 & 12 & 1 & 3 & -2 & 5 \end{array} \right] \\ \xrightarrow{R_3-2R_1} \left[\begin{array}{cccc ccc} 1 & 3 & -2 & 5 & 0 & 2 & -6 & 4 \\ 1 & 5 & -8 & 9 & 2 & 4 & 5 & 12 \\ 0 & -2 & 9 & 2 & 0 & 0 & 3 & 6 \end{array} \right] \\ \xrightarrow{R_3+R_2} \left[\begin{array}{cccc ccc} 1 & 3 & -2 & 5 & 0 & 2 & -6 & 4 \\ 1 & 5 & -8 & 9 & 2 & 4 & 5 & 12 \\ 0 & 0 & 3 & 6 & 0 & 0 & 3 & 6 \end{array} \right] \end{gathered}$		
johnston-linear-matrix-algebra-EQ1047	359	0.865 C +60	$\begin{aligned} & x_1 = \frac{\det \left(\begin{bmatrix} b_1 & a_{1,2} \\ b_2 & a_{2,2} \end{bmatrix} \right)}{\det \left(\begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \right)} = \frac{b_1 a_{2,2} - a_{1,2} b_2}{a_{1,1} a_{2,2} - a_{1,2} a_{2,1}} \text{ and} \\ & x_2 = \frac{\det \left(\begin{bmatrix} a_{1,1} & b_1 \\ a_{2,1} & b_2 \end{bmatrix} \right)}{\det \left(\begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \right)} = \frac{a_{1,1} b_2 - b_1 a_{2,1}}{a_{1,1} a_{2,2} - a_{1,2} a_{2,1}}. \end{aligned}$		

Conf.

MathPix confidence:

■

≥0.9

■

0.5-0.9

■

<0.5

 — no value

LaTeX source

MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual

render vs scan (inkdrill):

●

C component

●

W weak

●

S stable

●

N noise

●

K clean

●

not measured

[refined: *basis*]

beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.